Chirag Patil

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Education

Pune Institute of Computer Technology (PICT) — Pune, Maharashtra, India
Bachelor of Technology (B.Tech), Electronics and Telecommunication Engineering — *2024 – 2028*

Indian Institute of Technology Madras (IIT Madras)
Bachelor of Science (B.Sc), Data Science and Programming — *2024 – 2028*

Technical Skills

Languages: Java, C++, SQL
Frameworks: Spring Boot, Spring Framework, Spring Security, Servlets, JPA, FastAPI
Databases: MySQL, PostgreSQL
Cloud & Tools: AWS S3, AWS RDS, Git, GitHub, Maven, Gradle, IntelliJ IDEA, Android Studio, Postman

Projects

SAR Generator with Audit Trails
Spring Boot, Java, PostgreSQL, AWS S3, JWT, Python (AI Microservice) — Hackathon Project

- Built a Spring Boot backend for automated Suspicious Activity Report (SAR) generation with RESTful APIs, JWT-based authentication, and Role-Based Access Control (RBAC) for Analyst, Admin, and Auditor roles.
- Designed a scalable data ingestion pipeline for CSV and JSON files, integrating AWS S3 for secure storage and managing 15+ relational entities using PostgreSQL and JPA (Hibernate).
- Integrated AI-powered analysis via Python microservice for SAR narrative generation, typology detection, and end-to-end compliance workflows; implemented audit logging and DTO-based architecture for data integrity.

ValidEDI – Healthcare EDI Parser & Validator
FastAPI, PostgreSQL, SQLAlchemy, Firebase, AWS S3, Groq LLM — Hackathon Project

- Built an async FastAPI platform to process X12 HIPAA EDI transactions (837/835/834) with custom parsing, loop hierarchy construction, and 50+ validation rules.
- Implemented Firebase JWT-based RBAC with 6 roles; integrated AWS S3 and PostgreSQL (JSONB, async ORM) for scalable storage and batch processing, with AI-powered EDI analysis via Groq Llama 3.3.

ARTHIX – Zero-Typing UPI Expense Tracker
Kotlin, Jetpack Compose, Room + SQLCipher, Hilt, ML Kit, Whisper (ONNX) — iQOO Hackathon 2026 – FinTech & Commerce

- Built a fully on-device Android expense tracker that logs UPI spends with no typing: bank/UPI notifications (`NotificationListenerService`) supply the exact amount and payee, while a phone shake supplies the category.
- Designed a nearest-neighbour reconciliation engine (120s symmetric window) pairing shakes and notifications in either order; all amounts stored as `Long` paise to avoid floating-point errors.
- Privacy-first design: no backend, AES-256 SQLCipher-encrypted Room DB, bounded accelerometer window for battery; LLM-phrased reports validated against deterministic math, plus offline Whisper voice splitting and ML Kit OCR.

Achievements and Activities

Barclays Summer Intern (Upcoming): Selected for Barclays' Internship Program after clearing the Hack-O-Hire hackathon round and subsequent interview process.

Hackathons: Winner at COEP Inspiron; Top 7 at iQOO Hackathon; Finalist at Barclays Hack-O-Hire, Tata Motors COEP MindSpark '25, and PISB Techrush 2025.

NDA & SSB: Cleared the National Defence Academy (NDA) written examination (UPSC) and Stage 1 of the Services Selection Board (SSB).

National-Level Debater: Represented institution at national-level debate competitions.

Member – Computer Society of India (CSI), PICT Pune: Active in coding contests and workshops.